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Concept drift detection in image data stream: a survey on current literature, limitations and future directions

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Abstract

Concept drift—changes in the underlying data distribution over time—poses a significant challenge to machine learning systems deployed in dynamic environments. While numerous drift detection methods have been developed for structured data such as tabular and time-series streams, concept drift in image data remains an underexplored area due to the unstructured and high-dimensional nature of visual information. This survey presents the first comprehensive review of concept drift detection methods tailored for image data streams. We propose a novel taxonomy that categorizes existing approaches based on key properties such as image feature handling, detection strategy, detection level, concept drift cause, and evaluation considerations. Through the lens of this taxonomy, we analyze 14 representative concept drift detection methods designed for image data, highlighting current approaches to the field, their strengths and limitations. Based on this analysis, we outline promising future research directions to advance the field of concept drift detection in image-based systems.

Keywords Concept drift detection · Image data streams · Deep learning · Image representation learning

Abbreviations

ACC	Accuracy
ADWIN	Adaptive Windowing
AUROC	Area Under the Receiver Operating Characteristic Curve

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CD	Concept Drift
CDD	Concept Drift Detection
CNNs	Convolutional Neural Networks
DA	Domain Adaptation
DDD	Data Drift Detection
DDM	Drift Detection Method
ECE	Expected Calibration Error
FA	False Alarm
FPR95	False Positive Rate at 95% True Positive Rate
GCD	Generalized Category Discovery
GMM	Gaussian Mixture Model
GNNs	Graph Neural Networks
GSD	Gaussian Split Detector
GSP	Graph Signal Processing
IKS	Incremental Kolmogorov-Smirnov
KL	Kullback–Leibler Divergence
KS	Kolmogorov-Smirnov
LSDD	Least Squares Density Difference
MAD	Median Absolute Deviation
ME	Mean Embedding
MRMR	Minimum Redundancy Maximum Relevance
MSD	Mean Squared Deviation
ND	Novelty Detection
OOD	Out-of-Distribution
OSR	Open Set Recognition
PCA	Principal Component Analysis
SCF	Smooth Characteristic Function
STUDD	Student-Teacher Unsupervised Drift Detection
SVD	Singular Value Decomposition
UDD	Uncertainty Drift Detection
UMAP	Uniform Manifold Approximation and Projection
ViTs	Vision Transformers

1 Introduction

1.1 Motivation

In dynamic environments, changes are inevitable, and the systems we build often need to account for these shifts. From abnormal weather patterns affecting climate models to market trends affecting stock trading, many real-world processes are subject to changes that occur over time. These changes can happen unpredictably, and they pose a significant challenge to systems designed to make accurate predictions based on historical data. When the underlying characteristics of data evolve, models that previously worked well begin to fall short of accurate predictions—a phenomenon known as concept drift.

Concept drift, first introduced by Schlimmer and Granger (1986), refers to changes in the conditional distribution $P(y|x)$, where the relationship between input features x and target labels y evolves over time in non-stationary environments. In the other hand, data drift (also known as covariate shift) denotes changes in the input features distribution $P(x)$ (Ackerman et al. 2020). For example, a change in lighting conditions in an image stream may constitute data drift, whereas the emergence of new object classes or evolving definitions of existing classes represents concept drift. Data drift may or may not result in concept drift; however, it is often considered an early indicator of concept drift and is commonly addressed in concept drift as virtual drift. A more detailed discussion of this distinction is provided in Sect. 2.

Concept drift is a central problem in dynamic learning scenarios, where data distributions evolve over time. Beyond this setting, several related research areas have emerged, including domain adaptation, out-of-distribution detection, novelty detection, open set recognition, and generalized category discovery. While not all of these explicitly assume temporal or streaming data, they share conceptual similarities with concept drift and are often discussed together. However, their boundaries are sometimes blurred in the literature, leading to inconsistent usage and overlapping terminology. To provide clarity, we attempt to make a comparison between these concepts in Table 1. A key distinction among these fields lies in their primary focus. Concept drift detection and data drift detection address the temporal

Table 1 Comparison of concept drift with related fields

Concept	Definition	Main domain and applications
Concept Drift Detection (CDD) (Gama et al. 2014)	Change over time in the conditional distribution $P(y\|x)$ (i.e., relationship between features and labels)	Streaming and dynamic learning, with applications in remote sensing, medical imaging, network intrusion, fraud detection, ...
Data Drift Detection (DDD) (Ackerman et al. 2020)	Change in the marginal distribution of inputs $P(x)$ without affecting label distribution	Model monitoring and data quality assurance, with applications in machine learning pipelines, recommender systems, computer vision, ...
Out-of-Distribution (OOD) Detection (Yang et al. 2024)	Identifying test samples that lie outside the support of training distribution	Robustness and safety in machine learning models, with applications in autonomous driving, computer vision, natural language processing, ...
Novelty Detection (ND) (Yang et al. 2024; Pimentel et al. 2014)	Detecting the emergence of entirely new, previously unseen classes. In the context of CDD, ND can be used to discover new classes	New class discovery, with applications in cyber security, computer vision, medical imaging, ...
Open Set Recognition (OSR) (Yang et al. 2024; Geng et al. 2021)	A subclass of ND, OSR trains a classifier to handle both known classes and reject unknowns at test time	New class discovery, with applications mainly in the computer vision and natural language processing field
Generalized Category Discovery (GCD) (Vaze et al. 2022)	Discovering novel categories in unlabeled test data while also recognizing known categories. GCD goes beyond OSR by not only rejecting but also clustering/labeling unknowns into meaningful new classes	New class discovery, with applications mainly in the computer vision and natural language processing field
Domain Adaptation (DA) (Farahani et al. 2021)	Transferring knowledge from a source domain to a target domain with distribution shift	Cross-domain generalization, with applications in computer vision, medical imaging, speech recognition, ...

dimension, as they monitor how distributions evolve over time (Gama et al. 2014; Ackerman et al. 2020) in dynamic or streaming environments. In contrast, out-of-distribution detection, open set recognition, novelty detection, and generalized category discovery are not inherently temporal problems; instead, they operate by comparing data encountered at test time with what was observed during training, with the aim of identifying whether samples belong to unknown or unseen distributions or classes (Yang et al. 2024; Pimentel et al. 2014; Geng et al. 2021; Vaze et al. 2022). However, approaches in these fields can still be applied to concept drift or data drift problems: for example, novelty detection techniques can help identify emerging classes in real drift scenarios, while out-of-distribution detection methods may support early warning signals of distributional changes. Domain adaptation, on the other hand, differs from both groups: rather than detecting shifts, its goal is to adapt models trained on a source domain to perform effectively on a target domain with different statistical properties (Farahani et al. 2021). This distinction highlights why concept drift requires specialized methods, since its challenges are inherently tied to evolving data streams and the necessity for continual adaptation.

In the context of image data, concept drift poses critical challenges that directly impact the reliability of computer vision systems. For instance, an image classification model deployed in autonomous vehicles may face evolving conditions such as new weather patterns, varying lighting, or unfamiliar road signs, all of which alter the relationship between visual features and semantic labels. Similarly, in medical imaging, the manifestation of diseases may change over time or new imaging technologies may be introduced, leading to shifts in how features correspond to diagnostic categories. In both scenarios, the conditional distribution $P(y|x)$ evolves, rendering the model's previously learned parameters suboptimal. These examples illustrate why concept drift is particularly important in image-based applications: visual data are inherently dynamic, and without mechanisms to detect and adapt to drift, models risk significant degradation in performance and reliability in real-world deployments.

Concept drift detection has attracted growing attention over the past decades. In its early stages, most existing methods relied heavily on ground-truth labels to detect changes in data distribution (Gama et al. 2004; Gama and Castillo 2006; Baena-Garcıa et al. 2006). More recently, research interest has shifted toward unsupervised approaches that aim to detect drift without requiring labeled data. However, the majority of these techniques are designed for structured (tabular) or time-series data and often perform poorly when applied directly to image data. A common strategy to address this gap involves extracting feature embeddings from images and applying concept drift detection methods to the resulting representations. The advent of deep learning in the 2010 s has revolutionized image analysis, particularly in feature extraction, with models such as Convolutional Neural Networks (CNNs) (Krizhevsky et al. 2012) and Vision Transformers (ViTs) (Dosovitskiy et al. 2021) demonstrating remarkable capabilities in learning rich image representations. Leveraging deep learning models is therefore critical for improving feature extraction quality, which in turn enhances the accuracy and robustness of drift detection in image data. This survey focuses not only on traditional approaches but also emphasizes deep learning-based image representations, recognizing their crucial role in effectively handling image data and improving concept drift detection accuracy.

The existing literature surveys on concept drift are extensive but vary in focus. We summarize existing surveys in Table 2 to point out their strengths and limitations, as we high-

Table 2 Summary of existing survey papers on concept drift

Survey paper	Year	Data type	Strengths	Limitations
A survey on concept drift adaptation (Gama et al. 2014)	2014	Tabular	Provide a foundation knowledge on concept drift and adaptation strategies	Predates the era of deep learning and no mention of image data
A comparative study on concept drift detectors (Gonçalves et al. 2014)	2014	Tabular	A thorough comparison of traditional concept drift detectors	Predates the era of deep learning and no mention of image data
Learning under Concept Drift: A Review (Lu et al. 2019)	2018	Tabular	Propose a three-component framework for concept drift adaptation and summarize popular synthesis and real-world datasets	Emphasizes traditional ML approaches but lack mention of deep feature learning for images
Concept drift adaptation methods under the deep learning framework: A literature review (Xiang et al. 2023)	2023	Tabular	Categorizes deep learning-based methods and discussions on update and detection modes	Focus mainly on tabular data and lack mention of feature learning for image data
A benchmark and survey of fully unsupervised concept drift detectors on real-world data streams (Lukats et al. 2024)	2024	Tabular	A comprehensive benchmark of fully unsupervised detection methods with in-depth analysis	Focus on fully unsupervised methods only and lack coverage for image data

light the importance of our survey paper. Gama and Gonalves' surveys (Gama et al. 2014; Gonçalves et al. 2014) offer foundational knowledge about concept drift and detection/adaptation strategies, yet they predate the era of deep learning dominance and lack focus on unstructured data like images. Lu's review (2019) expands on detection methods and real-world applications but still underrepresents deep learning approaches, especially in handling concept drift in image data. Xiang's paper (2023) focuses on concept drift adaptation in deep learning models, offering insights into methods like incremental and transfer learning. However, it does not address the challenges specific to image data or representation learning. Recently, Lukats' survey (Lukats et al. 2024) gives a comprehensive benchmark on fully unsupervised concept drift detectors, which is the trend in concept drift detection, yet most of the mentioned papers are designed for tabular data only.

In contrast to previous survey papers, our work seeks to make a distinctive contribution to the research literature through the following goals:

- Propose a dedicated taxonomy that captures the key characteristics and properties of concept drift detection methods specifically for image data.
- Provide an overview of traditional concept drift detection methods and a critical analysis of image-based concept drift detection methods to highlight their strengths and limitations, thereby identifying significant gaps and underexplored areas in the current literature.
- Outline promising future research directions aimed at addressing these gaps and advancing the field of concept drift detection for image data.

1.2 Survey structure

The rest of this survey is organized as follows: Sect. 2 introduces the fundamental background on concept drift. In Sect. 3, we review traditional concept drift detection approaches and summarize the current challenges in traditional methods. Section 4 proposes a taxonomy of key properties for concept drift detection methods in image data. Section 5 presents our discussion of the surveyed methods, their strengths and weaknesses through the lens of the proposed taxonomy. Section 6 discusses open challenges and future research directions. An overview of the survey structure is illustrated in Fig. 1.

2 Background

2.1 Definition of concept drift

The term concept drift was first introduced by Schlimmer and Granger (1986) in his work on incremental learning algorithms, where he addressed the challenge of adapting to changes in the underlying data distribution over time. Since this publication, more researchers have expanded their point of view on the concept, most of them refer to the fact that the underlying data stream distribution changes over time (Widmer and Kubat 1996; Iwashita and Papa 2019; Desale and Shinde 2022). Concept drift has also been referred to by different authors under alternative terms, such as dataset shift (Quiñonero-Candela et al. 2009) or concept shift (Widmer and Kubat 1996).

In this section, we will define concept drift in line with the most commonly accepted interpretation found across the majority of the literature (Gama et al. 2014; Lu et al. 2019; Xiang et al. 2023; Greco et al. 2024; Kuppa and Le-Khac 2022). “Concept" refers to the underlying relationship between the input features X and the target outputs Y. This “concept" can be expressed as the joint probability distribution P, which shows how likely certain values of X and Y occur together. According to the chain rule of probability, we can represent this joint probability distribution mathematically as follows:

$$P(X, Y) = P(X) \cdot P(Y|X) \tag{1}$$

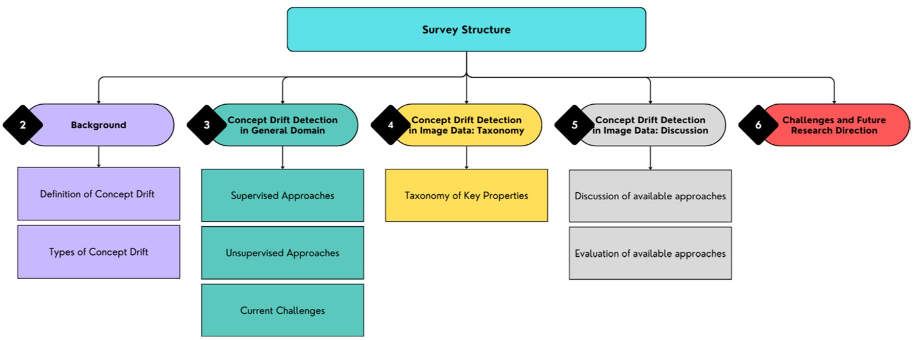

Fig. 1 Overview of the survey

Concept drift occurs when the joint probability distribution changes over time. In other words, there is a point in time t_1 that differs from the initial time t_0 where the distribution P has shifted. This shift can be formally expressed as:

$$\exists X : P_{t_0}(X,Y) \neq P_{t_1}(X,Y) \tag{2}$$

As in Eq. 1, we can understand that the causes for the change in distribution P could either come from the input distribution $P(X)$, or the conditional output distribution $P(Y|X)$, or sometimes both. Hence, we can distinguish different causes of concept drift:

1. *Virtual drift* (Delany et al. 2005) refers to changes in the distribution of input features without altering the conditional distribution of outputs, i.e., $P_{t_0}(X) \neq P_{t_1}(X)$ while $P_{t_0}(Y|X) = P_{t_1}(Y|X)$. Unlike data drift, which broadly denotes shifts in the input distribution, virtual drift is defined more strictly by its lack of impact on the decision boundary. Nonetheless, virtual drift is often regarded as an early indicator of real concept drift, and in practice, both types frequently occur simultaneously.
2. *Real concept drift* refers to the changes in the conditional distribution of outputs, i.e. $P_{t_0}(Y|X) \neq P_{t_1}(Y|X)$. These changes can happen with or without virtual drift and can result in a change in the decision boundary. To fully understand the causes of real concept drift, we recall the Bayesian Decision Theory (Duda et al. 2001):

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)} \tag{3}$$

   According to the equation, we can see that $P(Y|X)$ can be affected by two components: $P(Y)$, the prior distribution of outputs, and $P(X|Y)$, the likelihood of input features given a particular output. These components, when studied individually, are often treated as separate machine learning problems, highlighting the complexity behind real concept drift. Changes in $P(Y)$ together with $P(X)$ could imply the problem of novelty classes, while shifts in $P(X|Y)$ could suggest an in-class evolution of input features, meaning that some associated patterns/characteristics between X and Y have been changed. Real concept drift has also been referred to as concept shift (Salganicoff 1997) and conditional change (Gao et al. 2007).

In practice, most concept drifts are mixtures of virtual drift and real concept drift. This means that changes in the input distribution $P(X)$ and the conditional distribution $P(Y|X)$ often occur simultaneously, creating a more complex drift scenario. Figure 2 provides an intuitive illustration of the different causes of concept drift in a two-dimensional vector space, demonstrating how each type of drift can influence the decision boundary. This visualization helps clarify the distinct impacts of virtual drift (which does not affect the decision boundary) and real concept drift (which can cause significant changes to the decision boundary).

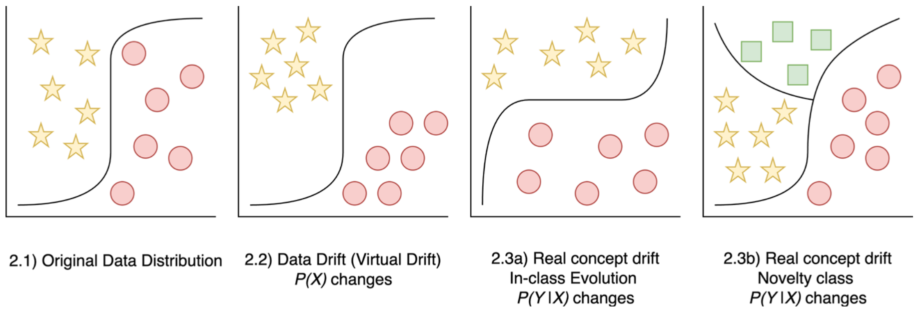

Fig. 2 Visualization of different causes of concept drift

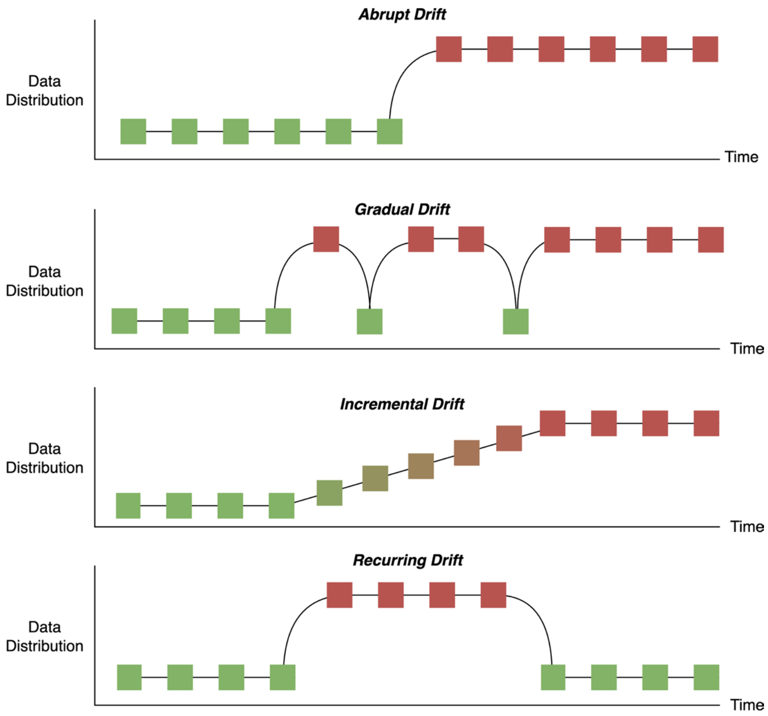

Fig. 3 Visualization of different types of concept drift

2.2 Types of concept drift

Concept drift can appear in different forms, depending on how the underlying data distribution changes over time. According to Gama et al. (2014), there are four most common types (illustrated as in Fig. 3): *Abrupt drift* occurs when there is a sudden shift from one concept to another; *Incremental drift* happens more gradually, as the data slowly transitions from one concept to another; *Gradual drift* refers to a situation where the old and new concepts coexist for a time before one eventually dominates; *Recurring drift* occurs when a previously seen concept reappears after some time.

To illustrate these different types, let's consider an example of an autonomous vehicle system tasked with detecting pedestrians and objects on the road:

- *Abrupt Drift*: The vehicle might be deployed in a new environment, such as moving from a city with clear, well-marked roads and traffic signs to a rural area with dirt paths and unclear traffic signs. This sudden change in the environment can cause an abrupt shift in the types of objects the model detects, making its previous training inadequate.
- *Incremental Drift*: Over time, the lighting conditions change gradually throughout the day, from bright sunlight to dusk and eventually nighttime. As the lighting changes incrementally, the model's ability to detect pedestrians and objects slowly shifts, requiring gradual adaptation to different lighting conditions.
- *Gradual Drift*: Imagine that the vehicle operates in an environment where road conditions and weather fluctuate. For a few hours, it might be raining, affecting the visual clarity of the sensors. The model might need to oscillate between handling dry roads and wet roads before eventually stabilizing when the rain stops. The gradual transition between these conditions leads to a drift in the model's performance.
- *Recurring Drift*: In this case, the vehicle may experience different driving conditions depending on the season. For example, in winter, it has to deal with snow-covered roads, while in summer, it encounters clear roads. This seasonal recurrence requires the model to adapt to the recurring changes in the environment, where the same road looks different depending on the time of year.

Most adaptive learning techniques, whether explicitly designed or implicitly structured, tend to focus on handling specific types of concept drift, with a heavy emphasis on sudden, nonrecurring changes. These abrupt drifts, where the data distribution shifts dramatically and immediately, are relatively easier to detect and address with traditional methods such as the Drift Detection Method (DDM) (Agrahari and Singh 2022). However, in real-world applications, concept drift is rarely isolated to a single type; it is often a mixture of various drift types such as gradual, incremental, or recurring drift, which adds complexity to detection and adaptation.

2.3 Related concepts

2.3.1 Self-supervised learning in concept drift

Self-supervised learning (SSL) has emerged as a powerful paradigm in representation learning, particularly in computer vision and natural language processing. Unlike supervised learning, which requires large amounts of labeled data, SSL leverages pretext tasks or self-generated supervision (e.g., predicting missing parts of an input, contrastive similarity learning) to learn meaningful feature representations directly from unlabeled data. Methods such as SimCLR (Chen et al. 2020), MoCo (He et al. 2020), BYOL (Grill et al. 2020), SwAV (Caron et al. 2020), and masked autoencoders (MAE) (He et al. 2022) have demonstrated that SSL can produce embeddings that are both semantically rich and robust to noise in the input features.

These properties are particularly relevant in the context of concept drift. First, SSL representations reduce sensitivity to input variations, thereby lowering the risk of false alarms due to virtual drift while still highlighting real drift that affects the conditional distribution $P(y|x)$. Second, since labels are often scarce or delayed in streaming environments, SSL offers a natural way to build or maintain robust feature spaces for drift detection without continuous reliance on labeled data. Finally, SSL has shown promise in related areas such as novelty detection and open-world recognition (Tack et al. 2020; Cao et al. 2022; Vaze et al. 2022), where the ability to separate known from unknown classes directly supports the detection of real drift scenarios like the emergence of novel classes. Taken together, SSL provides both practical tools and conceptual alignment with the challenges of concept drift, making it an important direction for future research in non-stationary learning.

2.3.2 Concept drift interpretability and adaptation

Beyond the task of identifying whether drift has occurred, an important dimension of research is drift interpretability, which seeks to understand, explain, and localize the changes driving the drift. In this sense, drift detection can be viewed as the first step of interpretability: methods such as DDM (Gama et al. 2004), EDDM (Baena-Garcıa et al. 2006), and ADWIN (Bifet and Gavalda 2007) not only signal the presence of drift but also provide indirect information about its nature, for example whether it is sudden or gradual. More advanced approaches go further by attributing drift to particular features, instances, or subspaces of the data. For instance, CDLEEDS (Ikonomovska et al. 2010) use tree-based models to highlight which parts of the input space are most affected, while statistical and distributional tests (Gretton et al. 2012) or feature-responsibility analysis (Rabanser et al. 2019) aim to interpret which variables are responsible for observed changes. Such interpretability is crucial for distinguishing between virtual and real concept drift, and for informing how adaptation should proceed.

Adaptation represents the next step after drift detection and interpretation, focusing on how models should evolve once a change has been identified. In details, adaptation refers to updating or reconfiguring models so that they can maintain predictive accuracy under non-stationary conditions. A wide range of strategies have been developed to achieve this. Classical approaches include instance selection or weighting (Gama et al. 2014), sliding windows, and ensemble-based methods such as DWM (Kolter and Maloof 2007), Learn++.NSE (Elwell and Polikar 2011), and Adaptive Random Forests (Gomes et al. 2017). Online learners like Hoeffding Trees (Domingos and Hulten 2000; Bifet and Gavaldà 2009) incrementally update their structure as new data arrive, ensuring responsiveness to evolving distributions. More recently, meta-learning and transfer learning techniques (Moradi et al. 2024; Yu et al. 2022; McKay et al. 2020) have been introduced to accelerate adaptation, particularly in scenarios where concepts recur or related drifts appear across domains.

Together, drift interpretability and adaptation form an integrated pipeline for continuous learning: drift detection delivers the initial signal, interpretability reveals what has changed and why, and adaptation guides how the model should evolve in response. Incorporating all three layers ensures that systems are not only resilient to non-stationarity but also transparent and trustworthy in how they react to evolving data.

3 Traditional concept drift detection methods

This section provides an overview of general concept drift detection methods, including popular and classic approaches widely recognized in the literature. While these methods may not be specifically designed for or tested with image data, they offer foundational principles and techniques for addressing concept drift across various domains. Besides, this section introduces how drift detection methods are typically classified in the literature. This classification serves as a framework for understanding the diverse methodologies used to tackle concept drift, helping to contextualize the more specialized approaches discussed later in this survey.

In the literature, many survey papers have addressed concept drift detection, but different papers categorize concept drift detectors into different groups. Several surveys classify concept drift detection methods based on whether they operate in a supervised or unsupervised setting (Hinder et al. 2024; Khamassi et al. 2018). The supervised methods rely on ground-truth labels, while unsupervised methods operate when labeled data is not available or limited. Semi-supervised methods can also work in an unsupervised setting since they only use labeled data in the training phase. In this survey, we consider semi-supervised methods as unsupervised methods.

Another common categorization divides concept drift detection methods into performance/error rate-based, distribution-based, and multiple hypothesis-based categories (Lu et al. 2019; Bayram et al. 2022). These categories are not specifically tied to whether the methods are supervised or unsupervised. However, when reviewing the methods within each category, it becomes evident that supervised approaches often focus on monitoring statistical measures or error rates. In contrast, unsupervised methods frequently utilize distribution-based techniques and hypothesis tests.

3.1 Supervised methods

Supervised methods in concept drift detection are utilized in scenarios where labeled data is available or can be obtained incrementally. These methods typically monitor the performance of a learning model, such as error rate or statistical measures like accuracy, precision, recall, etc. to identify significant deviations that indicate a drift in the underlying data distribution. By leveraging labels, supervised drift detection approaches can provide direct and interpretable feedback on the impact of drift, making them highly effective for detecting abrupt or gradual shifts in data.

The classic Drift Detection Method (DDM) paper (Gama et al. 2004), proposed by Joao Gama in 2004, suggests monitoring the error rate of learning algorithms and detecting drift based on statistical differences in this error rate. It calculates a warning level and a drift level using a confidence interval approach and lets a new model learn when the drift level is reached. DDM's simplicity and effectiveness in detecting abrupt drifts laid the foundation for several later methods, each addressing specific limitations of the original technique. One key limitation of DDM is its sensitivity to long-term gradual changes, where drifts occur slowly over time, making it less responsive to such scenarios. To address this, the paper Early Drift Detection Method (EDDM) (Baena-Garcıa et al. 2006) extends DDM by considering the distance between two errors rather than just the error rate itself. Learning with Local Drift Detection (LLDD) (Gama and Castillo 2006) extends DDM by focusing on detecting drifts in local regions of the instance space rather than monitoring the entire model globally. Further developments like the Hoeffding Drift Detection Method (HDDM) (Frias-Blanco et al. 2014) and the Reactive Drift Detection Method (RDDM) (Barros et al. 2017) were also built on the foundations laid by DDM. RDDM was designed to overcome the performance degradation of DDM in long stable concepts by periodically resetting its error calculation, making it more efficient for detecting drifts in extended, stable periods. HDDM, on the other hand, introduced the use of Hoeffding's inequality to detect significant changes in performance statistics, which allowed it to work in scenarios without assumptions about the data distribution. Also inspired by Hoeffding's inequality, Fast Hoeffding Drift Detection Method (FHDDM) (Pesaranghader and Viktor 2016) implements a sliding window to track the maximum and most recent probabilities of correct predictions, calculates the difference between these probabilities, and uses Hoeffding's inequality to set a threshold for this difference. FHDDM shows a balance between accurate drift detection and reduced computational complexity.

The ADWIN (Adaptive Windowing) algorithm, proposed by Bifet and Gavalda (2007), represents a significant advance in concept drift detection by automatically adjusting the window size based on the rate of change in the data stream. Unlike fixed-window techniques, ADWIN continuously monitors the statistical properties of the data within its window, dynamically shrinking the window when a change is detected and expanding it when the data remains stable. ADWIN's extension, ADWIN2 (proposed in the same paper), further improves time and memory efficiency by utilizing data stream algorithmics, allowing it to scale for large, real-time data streams with logarithmic memory and update time. This work laid the groundwork for subsequent research in adaptive windowing methods and has been widely applied in many real-world applications.

Other worth-mentioning methods in the field of concept drift detection include the Entropy-based Concept Shift Detection (ECSD) (Vorburger and Bernstein 2006), the Dynamic Extreme Learning Machine (DELM) (Xu and Wang 2017) and Statistical Test of Equal Proportions Drift Detection (STEPD) (Cabral and Barros 2018). ECSD leverages an entropy-based measure to detect concept shifts by comparing two sliding windows of data. Meanwhile, DELM introduces monitoring the Extreme Learning Machine (ELM) classifier's performance over time and raises an alert when the threshold is reached. STEPD detects concept drift by statistically comparing recent classifier accuracy with historical accuracy, signaling drift if the recent accuracy significantly decreases beyond a set threshold.

3.2 Unsupervised methods

Unsupervised concept drift detection methods play a crucial role in environments where labeled data is scarce or unavailable. These methods focus on identifying changes in the underlying data distribution when ground-truth labels are not available or limited. Unlike supervised approaches, which rely on tracking evaluation metrics, unsupervised methods monitor statistical properties of the data stream itself such as distributional changes to flag potential drifts. This makes them particularly useful in real-world scenarios where obtaining labeled data in real-time is costly and sometimes impractical.

Statistical tests are widely used in unsupervised concept drift detection due to their ability to identify changes in the underlying data distribution without relying on labeled data. Most of the works in the literature applied statistical tests to at least three different types of information, as discussed by Žliobaite (2010), including: (1) raw data, (2) classifier outputs, and (3) estimated class labels. Sobolewski and Woźniak (2013) conducted an empirical study comparing statistical tests for detecting virtual concept drift, specifically applying the Kolmogorov–Smirnov (KS) test, the Maximum Mean Discrepancy two-sample t-test (Gretton et al. 2012), and a few other tests to evaluate their effectiveness in different drift scenarios. Their approach involved testing each data feature individually to determine whether each sample came from the same population as a reference dataset. They concluded that the KS test and two-sample t-test performed best, with high specificity and sensitivity in detecting drift in both real and synthetic datasets. These findings demonstrated the efficacy of traditional statistical tests in concept drift detection, which laid a foundation for future work to refine and optimize these methods further. Later works have attempted to further improve the two-sample t-test by using binary classifiers (Lopez-Paz and Oquab 2017) or deep learning kernel (Liu et al. 2020), enabling more powerful drift detection in complex, high-dimensional datasets. dos Reis et al. (2016) proposed the Incremental Kolmogorov-Smirnov (IKS) algorithm, which enhances the traditional Kolmogorov-Smirnov test by enabling incremental updates, reducing computational costs and enabling real-time drift detection in streaming data. Some other worth-mentioning tests include the Least Squares Density Difference (LSDD) test (Bu et al. 2018) and the Mean Embedding (ME) test and the Smooth Characteristic Function (SCF) test (Jitkrittum et al. 2016).

Another common approach in unsupervised drift detection involves training a classifier to differentiate between the distributions of past and current data. Hido et al. (2008) first proposed the approach in their paper "Unsupervised change analysis using supervised learning", framing detecting change as a classification task. This approach deploys a classifier to discriminate between old and new distributions, using feature importance to highlight the primary drivers of drift. This supervised-insight approach for an unsupervised problem is versatile, offering explanatory insights alongside detection. Gozuacik further advanced this direction in two later works. In the first, he proposed a discriminative classifier approach (D3) (Gözüaçık et al. 2019) which monitors changes in the data distribution using a sliding window, detecting drift based on how well new samples align with previous feature distributions. In a later study, he proposed the one-class classifier (OCDD) (Gözüaçık and Can 2020) which focuses on outlier detection by identifying significant shifts in the data through unsupervised learning, making it ideal for scenarios with limited label access.

Other approaches focus on Model Uncertainty and Clustering. Baier et al. (2022) introduce Uncertainty Drift Detection (UDD), which detects drift by leveraging uncertainty estimates from neural network predictions, eliminating the need for labeled data. Using Monte Carlo dropout for Bayesian approximation, UDD identifies drift by monitoring changes in model uncertainty, thus minimizing unnecessary retraining on false alarms. Meanwhile, Chiu and Minku (2020) proposed clustering in the model space to manage an ensemble of models that are adapted to different concepts, enabling the detection and handling of both recurrent and novel drifts. The method maintains a memory of past models, where a clustering algorithm groups models based on prediction similarity, facilitating the retrieval of past knowledge when recurring drifts occur. Upon detecting drift, the framework creates a highly diverse ensemble from distinct clusters of past models, optimizing adaptation to drifts.

Recently, some novel approaches have been explored to detect concept drift more effectively. Cerqueira et al. (2023) present STUDD, a student-teacher model that identifies drift by tracking performance divergences between the student and teacher models over time. In this approach, the student aims to replicate the teacher's predictions, with drift flagged when discrepancies emerge, effectively using mimicking loss as a substitute for labeled feedback. Meanwhile, Gulcan and Can (2022) introduce the Label Dependency Drift Detector (LD3), tailored for multi-label data streams. This algorithm leverages label dependencies and ranking methods to detect drift by tracking shifts in label relationships, offering robustness in multi-label environments where traditional detection methods may fall short.

In another domain, graph-based representations have gained increasing attention as they naturally model entities and their complex interdependencies. Building on this foundation, Graph Neural Networks (GNNs) (Ponzi and Napoli 2025) and Graph Signal Processing (GSP) (da Rosa et al. 2022) have emerged as powerful paradigms within deep learning, enabling effective interpretation and knowledge extraction from graph-structured data. Within the context of concept drift detection, several works have explored unsupervised graph-based approaches. For instance, Seeliger et al. (2017) proposed a method that derives process graphs from event logs and monitors structural graph metrics over time; by applying statistical tests to these metrics, their approach can detect changes in process behavior without requiring labels. Similarly, Paudel and Eberle (2020) introduced the Discriminative Subgraph-based Drift Detector (DSDD), which identifies representative subgraphs in a graph stream and tracks the entropy of their distribution within a sliding window. Drift is then detected using direct density-ratio estimation to capture significant changes in subgraph distributions. Both methods highlight how graph-based modeling, even in the absence of supervision, can provide powerful tools for identifying distributional shifts in evolving data streams.

3.3 Discussion

Despite advancements in both supervised and unsupervised approaches, several key challenges persist in the literature:

- *Unsupervised Methods* The importance of unsupervised methods lies in their ability to address the missing labels problem. They offer a viable alternative for scenarios where labeled data is unavailable or difficult to obtain. However, in literature, supervised methods are still dominant in number (Lu et al. 2019), and designing robust unsupervised methods that can detect drift while maintaining low false positive rates remains a challenge.
- *Concept Drift Understanding* All drift detection methods are designed to detect "When" the drift happens, but very few methods have the ability to answer "How" and "Where".
- *Real-Time Detection* Fast and robust detection is essential for systems operating in dynamic environments. Achieving real-time performance without compromising accuracy remains a significant challenge, especially as data volumes grow.
- *Scalability* Handling large datasets efficiently and generalizing across multiple domains is critical for practical adoption. Existing methods often fail to scale to big datasets or adapt to domain-specific nuances, limiting their utility in real-world applications.
- *Drift Adaptation Speed* The speed at which a model responds to drift is crucial for minimizing performance degradation. Many methods suffer from delayed adaptation, which can be detrimental in time-sensitive applications like autonomous driving or real-time surveillance.

- *Evaluation Metrics* Evaluating drift detection remains inconsistent across the literature. Metrics for detecting drift are not standardized, making it difficult to compare methods. Developing comprehensive and domain-agnostic evaluation frameworks is an open problem.

To address these challenges, more integrated and domain-specific approaches are needed. Combining robust unsupervised drift detection with deep learning-based feature extraction shows promise for unstructured data like images. Additionally, developing scalable, real-time methods and standardizing evaluation metrics will be critical to advancing the field. By addressing these gaps, future research can enhance the applicability of concept drift detection methods across a wider range of domains and data types.

4 Concept drift detection in image data: a taxonomy of key properties

In this section, we will introduce our interpretation of taxonomy for concept drift detection in image data streams. Based on a search strategy, we have chosen from the literature relevant methods and summarize these methods together to propose a taxonomy of different characteristics/properties that are relevant when designing a concept drift detection method for image data.

4.1 Databases and search strategy

The literature search was conducted using a comprehensive selection of academic databases and digital libraries to ensure broad and relevant coverage of the topic. The sources include:

- *Google Scholar* (https://scholar.google.com) A freely accessible web search engine that indexes scholarly articles across a wide range of disciplines and publishers. It provides a broad overview of available literature, including peer-reviewed articles, theses, books, conference papers, and preprints.
- *Scopus* (https://www.scopus.com) A leading abstract and citation database of peer-reviewed literature, including scientific journals, books, and conference proceedings. Scopus offers extensive coverage of science, technology, medicine, social sciences, and arts and humanities, and is valued for its citation tracking and analytical tools.
- *IEEE Xplore* (https://ieeexplore.ieee.org) A digital library providing access to high-quality technical literature in electrical engineering, computer science, and electronics. It hosts publications from the Institute of Electrical and Electronics Engineers (IEEE) and other affiliated organizations, including journals, conference papers, standards, and eBooks.
- *ACM Digital Library* (https://dl.acm.org) A comprehensive collection of full-text articles and bibliographic records covering computing and information technology. Published by the Association for Computing Machinery (ACM), it includes journal articles, conference proceedings, technical magazines, and newsletters.
- *SpringerLink* (https://link.springer.com) A research platform that provides access to millions of scientific documents from journals, books, series, protocols, and reference works published by Springer. It is widely used in the fields of computer science, engi-

Table 3 Search keywords

Priority	Keyword	Synonyms
First	Concept Drift Detection	Concept Drift Detectors
Second	Virtual Concept Drift, Real Concept Drift	Covariate Dataset Shift, Semantic Dataset Shift
Third	Image Data	Image Dataset, High-Dimensional Data
Fourth	Novelty Detection, Open Set Recognition, Generalized Category Discovery	Novel Class Detection, Open Set detection

Table 4 Inclusion and exclusion criteria for literature selection

Inclusion criteria	Exclusion criteria
IC1: Articles published between 2018 and 2025	EC1: Articles published before 2018 or not peer-reviewed
IC2: Studies that propose or evaluate concept drift detection methods applied to image data	EC2: Studies focusing solely on concept drift in non-image data (e.g., tabular, text)
IC3: Papers presenting methodology that focus on either concept drift detection or image data handling	EC3: Papers that do not address either core concept drift detection problems or image data processing

neering, and applied sciences.

- *Elsevier/ScienceDirect* (https://www.sciencedirect.com) Elsevier is a major academic publisher, and ScienceDirect is its online platform offering access to a vast repository of scientific and technical research articles. It covers a wide range of disciplines, including computer science, engineering, health sciences, and the physical sciences.
- *arXiv* (https://arxiv.org) A preprint server for new research in physics, mathematics, computer science, quantitative biology, quantitative finance, statistics, electrical engineering, and systems science. Managed by Cornell University, arXiv is widely used for disseminating preliminary research results and open-access papers prior to peer review.

Advanced search features and Boolean operators were employed to effectively refine the search results across selected academic databases. The primary search keywords and their relevant synonyms are summarized in Table 3. These keywords were used in various combinations to ensure comprehensive coverage of the literature related to concept drift in image data.

To ensure relevance and quality of the selected works, we applied a three-stage filtering process based on the inclusion and exclusion criteria outlined in Table 4. An initial search yielded a total of 4,470 papers published between 2018 and 2025 that mentioned concept drift detection (meeting Criterion 1). Applying Criterion 2, which filters out studies that focus on non-image data types (e.g., tabular or textual data), reduced the pool to 271 papers. We then manually screened the title and abstract of each article to apply Exclusion Criterion 3, which removed studies that did not address either core concept drift detection techniques or image data processing. This final step resulted in a curated set of 11 papers selected for in-depth review and analysis.

Given the limited availability of dedicated concept drift detection methods for image data in the current literature, we broaden our scope to include approaches that address various causes of concept drift, such as virtual drift and real concept drift (as discussed in Section 2). To this end, we draw from related fields in computer vision, including Novelty Detection, Open Set Recognition and Generalized Category Discovery. Although these fields originate from different research setups, they share a common focus on handling distributional changes driven by the emergence of novel classes, an aspect closely aligned with the underlying causes of concept drift. We include three representative papers (Tack et al. 2020; Cao et al. 2022; Vaze et al. 2022) to highlight their primary focus and incorporate their approaches into our final taxonomy, as they offer valuable strategies that can be effectively applied to concept drift detection.

4.2 Taxonomy for concept drift detection in image data

Numerous studies have attempted to classify or provide a taxonomy for concept drift detection methods. Agrahari and Singh (2022) categorize detectors based on various criteria, while Lu et al. (2019) organize them according to the test statistics they employ. Additionally, Gemaque et al. (2020) introduce a taxonomy specifically for unsupervised approaches. However, these taxonomies are primarily designed for common concept drift detectors and do not account for the unique challenges posed by image data. Our proposed taxonomy does not aim to classify existing methods, but rather to outline and structure the key aspects and characteristics involved in designing concept drift detection methods for image data. In this section, we present our interpretation of the common properties and components that shape such methods. The proposed taxonomy is illustrated in Fig. 4.

4.2.1 Image features handling

Unlike other data types, images are unstructured and essentially just matrices of pixel values (e.g., RGB triplets). These numbers don't mean anything directly in terms of semantics, like shapes, objects, or scenes. Hence, to detect concept drift in image data, we need methods for handling these features. Image features handling often include the below categories:

- *Feature Extraction* Feature extraction aims to transform raw pixel-level information into structured, informative representations that capture essential patterns such as edges, textures, shapes, objects or semantic contexts.

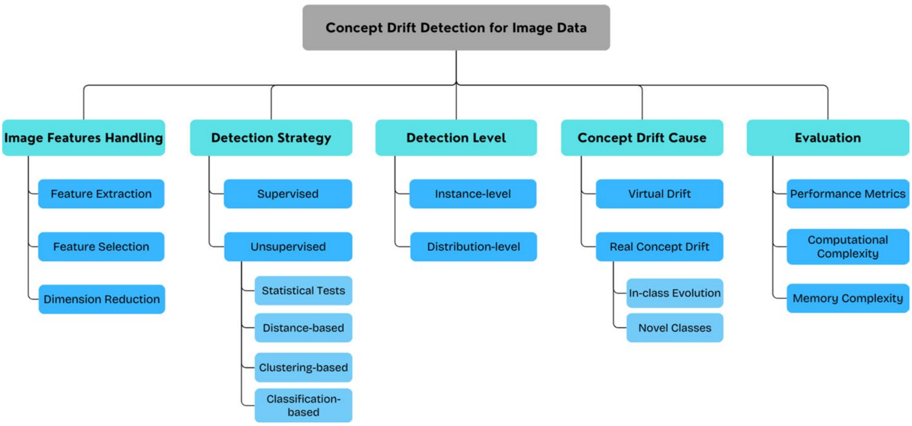

Fig. 4 Taxonomy of general characteristics of different concept drift detection approaches for image data

- *Feature Selection* Feature selection focuses on identifying the most relevant features while discarding redundant or noisy ones. Feature selection often comes after feature extraction when dealing with image data.
- *Dimension Reduction* Instead of selecting important features, dimension reduction refines the feature space by compressing it into a lower-dimensional manifold, preserving as much critical information as possible. This step not only reduces computational and memory requirements but also stabilizes statistical methods that are sensitive to high-dimensional noise.

4.2.2 Detection strategy

Building on the definitions introduced in Section 3, concept drift detection methods can be broadly divided into **supervised** and **unsupervised** approaches.

4.2.3 Detection level

Traditional concept drift detection methods have predominantly focused on distribution-level analysis, particularly within data stream mining. However, recent research trends propose finer-grained, instance-level detection, especially for complex and high-dimensional data such as images.

- *Distribution-Level* At the distribution level, methods aim to detect drifts between the overall data distributions across time. This can involve comparing statistical properties of batches or windows of data. Distribution-level detection is efficient for detecting global shifts and has better runtime efficiency.
- *Instance-Level* Instance-level detection focuses on identifying drift as individual samples. Methods operating at this level can detect subtle or localized drifts, such as the appearance of novel classes or in-class evolution. Instance-level approaches are important to pinpoint which instances directly cause the concept drift.

4.2.4 Concept drift cause

As outlined in Sect. 2, concept drift arises mainly from **virtual drift** (changes in the input distribution) and **real concept drift** (changes in the conditional distribution). Real concept drift can further be distinguished into **in-class evolution** and the emergence of **novel classes**.

4.2.5 Evaluation

- *Evaluation Metrics* The choice of evaluation metrics often depends on the primary focus of the drift detection method. For methods that emphasize accurate drift detection, evaluation typically involves drift-specific metrics such as detection delay, false alarm rate (false positives), and missed detection rate (false negatives). For methods where classification performance is also considered, standard machine learning metrics such as overall accuracy and the Area Under the Receiver Operating Characteristic curve

(AUROC) are often reported.

- *Computational Complexity* Computational complexity assesses the runtime efficiency of drift detection algorithms. For real-time applications, especially in high-dimensional image data streams, it is crucial that detection methods operate with low computational overhead to ensure timely responses to drift events.
- *Memory Complexity* Memory complexity measures the storage requirements of a method. Some algorithms require retaining large volumes of historical data or maintaining complex models of the feature space, which may not be feasible in resource-constrained settings.

5 Concept drift detection in image data: a comparison of current literature

In this section, we present 14 representative papers relevant to the task of concept drift detection. Among them, 11 originate from the traditional concept drift literature, while the remaining 3 are drawn from the computer vision domain, specifically addressing the problem of novel classes, one of the main causes of real concept drift. Although these papers operate under different assumptions and settings, we include them to illustrate promising research directions that can inform and inspire future work in concept drift detection for image data. We analyze these methods through the lens of our proposed taxonomy, highlighting their core characteristics and key distinctions. Each approach is first described in detail, followed by an evaluation of its effectiveness based on the criteria outlined in our framework.

5.1 Methods description

This section aims to present and characterize the key aspects of existing state-of-the-art approaches. Specifically, we summarize their core ideas and relevant features, and highlight their general properties, as organized in Table 5. Our analysis goes beyond the drift detection techniques themselves; we also examine how each method handles image feature representation, the specific causes of concept drift it addresses, and the datasets it employs. These dimensions are considered critical for effectively understanding and evaluating concept drift in image data.

5.1.1 High-dimensional multi-label data stream classification with concept drifting detection (MRMR)

Li et al. (2023) address the challenges of concept drift in high-dimensional multi-label data streams by using a feature selection-based framework. Their approach refines the minimal-redundancy-maximal-relevance (MRMR) feature selection method based on mutual information, and detects concept drift by monitoring changes in both the label distributions and the selected feature distributions between adjacent data chunks. Concept drift is detected if significant divergences are observed in either the label or feature spaces, using Hamming loss and cosine distance as detection criteria.

Table 5 Summary of concept drift detection methods for image data

Method	Image handling			Detection strategy	Detection level	Cause of drift			Datasets
	FE	FS	DR			VD	ICE	NC	
MRMR (Li et al. 2023)	✗	✓	✗	Supervised	Distribution	✓	✗	✗	Corel16k010 (Li and Wang 2008), NUS-WIDE (Chua et al. 2009)
CA-Drift (Cobb and Van Looveren 2022)	✓	✗	✗	Unsupervised—ST	Distribution	✗	✗	✓	ImageNet (Deng et al. 2009)
CDASC (Yuan et al. 2024)	✓	✗	✗	Unsupervised—ST	Distribution	✓	✗	✗	P.MNIST (Goodfellow et al. 2014), R.MNIST (Lopez-Paz and Ranzato 2017)
CDDBS (Okawa and Kobayashi 2021)	✓	✗	✗	Unsupervised—ST	Distribution	✓	✗	✗	CIFAR10 (Krizhevsky and Hinton 2009)
AMSC (Zhang et al. 2018)	✓	✗	✓	Unsupervised—DB	Distribution	✗	✗	✓	MNIST (Deng 2012)
DSM (Li et al. 2024)	✓	✓	✗	Unsupervised—DB	Distribution	✓	✗	✗	RSDDs (Gan et al. 2017)
DriftLens (Greco et al. 2024)	✓	✗	✓	Unsupervised—DB	Distribution	✓	✗	✓	Intel Image (Rahimzadeh et al. 2021), STL-10 (Coates et al. 2011)
CDCSDE (Xu and Klabjan 2021)	✓	✗	✗	Unsupervised—DB	Distribution	✓	✗	✗	MNIST (Deng 2012), USPS (Hull 1994)
IBDD (Souza et al. 2020)	✗	✗	✗	Unsupervised—DB	Distribution	✓	✗	✗	Ham10000 (Tschandl et al. 2018)
Image Drift (Fuccellaro et al. 2024)	✗	✗	✗	Unsupervised—DB	Distribution	✗	✓	✗	CIFAR10, CIFAR100 (Krizhevsky and Hinton 2009), MNIST (Deng 2012), Fashion-MNIST (Xiao et al. 2017)
Visual-Drift (Tran et al. 2024)	✓	✗	✓	Unsupervised—DB	Instance	✗	✗	✓	CIFAR10, CIFAR100 (Krizhevsky and Hinton 2009)

Table 5 (continued)

Method	Image handling			Detection strategy	Detection level	Cause of drift			Datasets
	FE	FS	DR			VD	ICE	NC	
CSI (Tack et al. 2020)	✓	✗	✗	Unsupervised—DB	Instance	✗	✗	✓	CIFAR10, CIFAR100 (Krizhevsky and Hinton 2009), ImageNet (Deng et al. 2009)
ORCA (Cao et al. 2022)	✓	✗	✗	Unsupervised—Classification-based	Instance	✗	✗	✓	CIFAR10, CIFAR100 (Krizhevsky and Hinton 2009), ImageNet (Deng et al. 2009), Single-cell (Almanzar et al. 2020)
GCD (Vaze et al. 2022)	✓	✗	✗	Unsupervised—Clustering-based	Instance	✗	✗	✓	CIFAR10, CIFAR100 (Krizhevsky and Hinton 2009), ImageNet (Deng et al. 2009)

FE: Feature Extraction, FS: Feature Selection, DR: Dimension Reduction, ST: Statistical Tests, DB: Distance-based, VD: Virtual Drift, ICE: In-Class Evolution, NC: Novelty Class

The method targets multi-label datasets such as Corel16k010 (Li and Wang 2008) and NUS-WIDE (Chua et al. 2009). There is no mention of the use of feature extraction, so we assume they operate on raw image pixels. This approach is supervised due to the need for ground-truth labels and works on the distribution level.

5.1.2 Context-aware drift detection (CA-Drift)

CA-Drift (Cobb and Van Looveren 2022) proposes a novel approach for concept drift detection inspired by causal inference. The key method builds upon the traditional Maximum-Mean-Discrepancy-based (MMD-based) two-sample test (Gretton et al. 2012), but introduces a context variable to allow differences in distribution. This allows the drift detector to take into account contexts that are expected to change and focus only on changes that cannot be attributed to the context variable. These context variables could be various factors that can lead to a variation in the distribution of data such as time of day, weather conditions, or any other domain-specific context. The method works by comparing the conditional distributions of the data, which is done by calculating conditional MMDs (based on kernel mean embeddings) between reference and deployment data, extended by the context variable.

The paper mentions experiments on the ImageNet dataset (Deng et al. 2009) using Santurkar's hierarchy classification (Santurkar et al. 2020) with 10 superclasses, each superclass containing 50 subclasses. They only used 25 subclasses as reference distribution for the superclass and the other 25 subclasses acted as a drifted alternative. The drifted samples come from different subclasses (new subclasses), hence, will be detected as they don't follow the same conditional distribution. This experiment utilized a pre-trained SimCLR (Chen et al. 2020) model for feature extraction, and its classification head for context.

5.1.3 Concept drift adaptation with scarce labels (CDASC)

CDASC (Yuan et al. 2024) proposes a mechanism to detect and adapt to concept drift in streaming data, leveraging diffusion learning when drift is detected and adversarial learning when no drift is observed. The approach integrates three key components: a feature extractor, a diffusion module, and two classifiers. The feature extractor is implemented as a two-layer fully connected neural network, while diffusion learning ensures that similar data points are mapped closer in the latent space, and dissimilar data points are further from each other. The method also addresses the challenge of scarce labels. It achieves this by mapping the features of unlabeled data into the latent space of labeled data, maximizing the discrepancy between the classifiers to identify data points near the decision boundary. These identified points are then used to adversarially update the feature extractor, effectively expanding the model's representation of the data distribution. To detect drift, the system continuously monitors the distance between the feature vectors of incoming data and the learned cluster centers. If a significant change is detected in the distance, a Kolmogorov-Smirnov (K-S) test is employed to verify whether concept drift has occurred.

CDASC is evaluated successfully on the Permuted MNIST (Goodfellow et al. 2014) and Rotated MNIST (Lopez-Paz and Ranzato 2017) datasets, which simulates virtual drifts of the original MNIST (Deng 2012) dataset. The approach demonstrates a new solution to concept drift detection and adaptation under conditions of limited labeled data, aligning closely with the objectives of unsupervised drift detection. However, the simplicity of the feature extractor poses potential challenges, including misrepresentation of complex data and scalability issues when applied to larger or more intricate datasets.

5.1.4 Concept drift detection method via boundary shrinking (CDDBS)

CDDBS (Okawa and Kobayashi 2021) employs an ensemble of different inspector models, each of which is trained to intentionally shrink within specific classification regions of its class. This shrinking ensures that the inspector models react more sensitively to changes in data distribution within their designated classes, making them highly responsive to concept drift. To train each inspector model, CDDBS removed data points that are mapped closely to the original model's decision boundary, or in other words, with low confidence scores. When drift occurs, the inspector models, due to their reduced boundary areas, are more likely to detect misclassified instances that have shifted within the classification space. The method uses classification scores obtained from the inspector models and calculates the Kullback–Leibler (KL) divergence with historical scores. If the divergence exceeds a predefined threshold, it signals the occurrence of drift in a specific class. The use of KL divergence helps to quantify how much the classification scores have deviated from their non-drifted state, thus providing a robust detection mechanism.

The authors proved the effectiveness of their method in detecting concept drift in image data using the CIFAR-10 dataset (Krizhevsky and Hinton 2009). Drift was synthetically simulated through various image manipulation techniques, such as blurring, darkening, rotation, etc. The method employs the MobileNetV2 (Sandler et al. 2018) architecture, a 53-layer deep convolutional neural network (CNN) optimized for mobile devices.

5.1.5 Adaptive matrix sketching and clustering (AMSC)

AMSC (Zhang et al. 2018) employs matrix sketching to model large data as a low-dimensional approximation matrix through singular value decomposition (SVD). Motivated by Mu et al. (2017), the method stores two kinds of matrix sketching: a global sketching G for the whole dataset and a local sketching L_i for every known class. The matrix G is updated dynamically whenever new data arrives to help identify both known and new classes from streaming data. The new class detection is done by calculating the distance between new instances with the matrix itself and comparing the distance with an adaptive threshold. The method also integrates the Clustering by Fast Searching and finding density peaks (CFS) algorithm (Rodriguez and Laio 2014) to effectively group data points into new clusters, minimizing the need for labeled data.

The author tested this approach on the MNIST dataset (Deng 2012), and the drift is simulated by reducing the number of initially known classes. For feature extraction, a CNN with two convolution layers is used and the output of the first layer is chosen as extracted features.

5.1.6 Label-free concept drift detection method based on a defect segmentation model (DSM)

DSM (Li et al. 2024) is a concept drift detection method tailored for industrial images. Unlike many approaches that rely on features extracted from the penultimate (pre-prediction) layer of a neural network, DSM adopts a feature selection algorithm called Relief (Urbanowicz et al. 2018). Specifically, the proposed method uses a neural network segmentation model to generate intermediate features and augments them with additional indicators such as grayscale and texture. The Relief algorithm then processes these combined features to produce a final, characteristic feature set. These features act as image representations, which are subsequently clustered to determine a central vector for each class. Each class has its own threshold, defined by the maximum distance of its in-class data points from the central vector. To quantify how new data points deviate from these clusters, the Mahalanobis distance is computed between incoming samples and each class's central vector; a data point is flagged as drift if it exceeds the class-specific threshold.

The effectiveness of this approach is demonstrated on the RSDDs-I and RSDDs-II datasets (Gan et al. 2017), both containing images of normal and heavy transport track surface defects. These datasets were selected because of their grayscale and texture characteristics, making the methodology particularly relevant.

5.1.7 Unsupervised concept drift detection from deep learning representations in real-time (DriftLens)

Greco et al. (2024) introduced DriftLens as an unsupervised real-time concept drift detection framework designed for unstructured data using deep-learning representations. DriftLens operates by extracting embeddings from a deep learning model and applying dimensionality reduction via Principal Component Analysis (PCA) (Maćkiewicz and Ratajczak 1993) in order to represent these embeddings as a multivariate normal distribution. The framework is composed of two phases: offline and online. In the offline phase, DriftLens constructs a baseline distribution of the deep learning embeddings derived from the initial training data. During the online phase, the framework processes new data in sliding windows and compares the distribution of embeddings from the new window to the baseline distribution using the Fréchet Distance (Dowson and Landau 1982) (also known as Wasserstein-2 distance). When this distance exceeds a predefined threshold, concept drift is detected, indicating that the new data deviates significantly from the previously observed distribution.

DriftLens is evaluated on the STL-10 (Coates et al. 2011) and Intel Image (Rahimzadeh et al. 2021) datasets, where concept drifts are simulated by (1) introducing new classes and (2) applying blur to the images. The framework leverages ViTs (Dosovitskiy et al. 2021) and VGG16 (Simonyan and Zisserman 2015) models to generate image embeddings.

5.1.8 Concept drift and covariate shift detection ensemble with lagged labels (CDCSDE)

CDCSDE (Xu and Klabjan 2021) proposes an ensemble-based approach to concept drift detection, combining six statistical measures to capture both real concept drift (changes in the relationship between features and labels) and virtual drift (changes in the feature distribution alone). The six metrics employed are: Exponentially Weighted Moving Average of classification error rates, Model uncertainty measurement, Hellinger distance between distributions, Auto-encoder reconstruction error, Sum-product network likelihood, and Natural gradient changes. Each metric generates a time series, and the method monitors these series to detect drift. Drift detection is achieved through a majority voting strategy across these metrics, as each contributes differently to the identification of drift.

The method is evaluated on image data using the MNIST (Deng 2012) and USPS (Hull 1994) datasets, both of which consist of handwritten digit images. Drift is simulated by training the model on MNIST and introducing USPS as the drift dataset due to differences in image quality and style. A 2-layer convolutional neural network is employed as the feature extractor.

5.1.9 Image-based drift detector (IBDD)

IBDD (Souza et al. 2020) is designed to address several key challenges in concept drift detection for streaming data, including limited label availability, real-time detection, and model dependence. The method operates by converting high-dimensional data from different time windows into two-dimensional visual representations, allowing for monitoring changes in data distribution over time. These visual representations are interpreted as grayscale images, where pixel intensity encodes the features of each image in the window. To measure similarity between the two images, IBDD employs the Mean Squared Deviation (MSD) metric. Drift detection is based on monitoring the time series of MSD values: if a sequence of values consistently exceeds or falls below a predefined threshold, concept drift is detected.

IBDD is evaluated on the Ham10000 dataset (Tschandl et al. 2018), a skin cancer dataset collected over 20 years. Drift is simulated by sorting the data ascending by the patient's age, reflecting the higher risk of skin cancer with age. IBDD works directly on raw image pixels without the use of any feature extraction.

5.1.10 Image drift

Fuccellaro et al. (2024) introduce the Gaussian Split Detector (GSD) to address the challenge of detecting concept drift in image data streams without requiring access to ground-truth labels during inference. Image Drift operates by modeling the distributions of selected features using Gaussian Mixture Models (GMMs) during training and estimating drifts in decision boundaries during inference. It computes optimal decision boundaries between classes based on Gaussian assumptions and flags drift when significant shifts in these boundaries are detected across an ensemble of simple splits. The method primarily detects real concept drift (changes that impact the decision boundary) and is explicitly designed to ignore virtual drift (distribution changes that do not affect classification).

The approach is evaluated on popular computer vision datasets, including MNIST (Deng 2012), Fashion-MNIST (Xiao et al. 2017), CIFAR-10, and CIFAR-100 (Krizhevsky and Hinton 2009), where drift is simulated through image perturbations such as Gaussian noise and geometric shifts. The method works on raw image pixels without using any feature extraction.

5.1.11 A new approach for concept drift detection in visual data (Visual-Drift)

Visual-Drift (Tran et al. 2024) introduces a pipeline method for detecting concept drift in image data. The pipeline consists of three key components: a feature extractor using a pre-trained transformer model, dimensionality reduction, and a distance-based algorithm for drift detection. Visual-Drift mentions the importance of selecting an embedding model capable of capturing the semantic meaning of images, ensuring effective representation in latent space. Dimensionality reduction is applied to lower computational costs while uncovering the underlying structure of the data. For drift detection, Visual-Drift utilizes Median Absolute Deviation (MAD) (Leys et al. 2013) to model the distribution of each class. This statistical measure accounts for variations in class tightness and calculates an adaptive threshold for each class. A drift is detected when a new data point falls outside the threshold of all known classes, identifying it as an outlier.

The pipeline employs CLIP (Radford et al. 2021) as the feature extractor and UMAP (McInnes et al. 2020) for dimensionality reduction. The method is tested on the CIFAR-10 and CIFAR-100 datasets (Krizhevsky and Hinton 2009), with drift simulated by introducing new classes from CIFAR-100 into the CIFAR-10 training set.

5.1.12 CSI: novelty detection via contrastive learning on distributionally shifted instances

CSI (Tack et al. 2020) addresses the challenge of novel classes for image data by proposing a new approach that leverages contrastive learning. CSI extends standard contrastive learning by introducing distribution-shifting augmentations (e.g., rotations, Gaussian noise) that simulate out-of-distribution samples. During training, CSI pushes these shifted augmentations away from original samples (treating them as negatives), while pulling together regular augmentations (positives), encouraging the network to learn features that can better separate in-distribution from shifted/out-of-distribution samples. For detection, CSI defines a new scoring function combining representation similarity and auxiliary classification on shifted instances to identify novel inputs.

The method targets image datasets including CIFAR-10, CIFAR-100 (Krizhevsky and Hinton 2009), and ImageNet (Deng et al. 2009), and is designed to handle scenarios where the novelty manifests as significant distributional shifts, similar to real concept drift through the introduction of new classes. CSI builds on contrastive representation learning (specifically SimCLR Chen et al. 2020) and uses a ResNet model as its backbone.

5.1.13 Open-world semi-supervised learning (ORCA)

ORCA (Cao et al. 2022) is introduced as a framework to address the open-world semi-supervised learning problem, where novel classes can appear in the unlabeled data at test time. ORCA operates by combining three main components: a supervised objective with an uncertainty adaptive margin, a pairwise objective for generating pseudo-labels among unlabeled data, and a regularization term to prevent trivial clustering solutions. First, ORCA introduces the uncertainty adaptive margin into the entropy loss to slow down the overfitting of seen classes. Secondly, to discover novel classes, ORCA includes a pairwise similarity learning task to generate pseudo-labels for new samples by identifying their nearest neighbors. These pseudo-labels are then used to group unlabeled data into clusters, which can discover novel classes. ORCA uses a set of classification heads, including extra heads that can be dynamically activated for novel class discovery based on clustering in the feature space. Finally, a regularization term using the Kullback-Leibler divergence to prevent the model from assigning all data into one class.

ORCA is evaluated on CIFAR-10, CIFAR-100 (Krizhevsky and Hinton 2009), ImageNet (Deng et al. 2009), and the Single-Cell Mouse Atlas dataset (Almanzar et al. 2020), targeting real concept drift where new classes emerge over time. The method uses deep feature extraction via a ResNet backbone pretrained with SimCLR (Chen et al. 2020) self-supervised learning. Features are continuously updated during ORCA's training, with additional novel class heads dynamically activated for discovered classes.

5.1.14 Generalized category discover (GCD)

Vaze et al. (2022) proposed the Generalized Category Discovery (GCD) framework to address a more realistic open-world learning scenario in which unlabeled data may contain both known and novel classes. The method detects novel classes through a combination of contrastive representation learning and semi-supervised clustering. First, GCD uses ViTs (Dosovitskiy et al. 2021) pretrained with DINO (Caron et al. 2021) and fine-tunes it using supervised and unsupervised contrastive losses, producing embeddings that preserve semantic similarity. After fine-tuning, the embeddings are clustered using a semi-supervised K-means algorithm, where centroids for known classes are initialized using labeled data and additional centroids are initialized for potential novel classes using K-means++ on unlabeled data. GCD runs k-means clustering with various values of k, and the optimal number of clusters is selected when labeled data clustering accuracy is maximized.

GCD is evaluated on both generic datasets (CIFAR-10, CIFAR-100 Krizhevsky and Hinton 2009, ImageNet Deng et al. 2009) and fine-grained benchmarks (SSB benchmark Vaze et al. 2022 and Herbarium19 Tan et al. 2019), focusing on real drift due to novel class emergence.

5.2 Methods analysis: evaluation metrics, quantitative results comparison, strengths and limitations

Based on the surveyed papers, we provide a comparison of concept drift detection methods, highlighting their respective strengths and weaknesses, as summarized in Table 6. In addition, we present quantitative comparisons of different methods in Table 7. Approaches that specifically address the detection of novel classes are discussed separately and summarized in Table 8.

5.2.1 Evaluation metrics

Across the literature, evaluation is tailored to the drift setting. For novel-class emergence, studies typically report standard classification metrics like Precision, Recall, F1, AUROC, and Accuracy, treating the appearance of a new class as a detection problem. For other drift types in streaming scenarios, many approaches augment Recall/F1/AUROC/Accuracy with event-aware and time-aware criteria. False alarms are often quantified in number or tracked as mean time between false alarms (MTFA), capturing stability against drift triggers. Detection Delay is another measurement, often referred to as the difference between when drift is detected and occurs. Other works also report Runtime/Latency and sometimes the number of Hidden Nodes (HN) in a neural network to establish feasibility under real-time or resource-constrained deployments.

Despite these common choices, evaluation in image-based, real-time streaming settings remains fragmented. First, metric definitions are inconsistent. For instance, "False alarms" may be reported as a raw count, a rate or mean time between false alarms (MTFA). Second, evaluation protocols vary widely. Crucial details like label latency, how many labels are available, and when the model is updated (immediately after an alarm or on a schedule) are often unstated, yet they directly influence the metrics results. Reporting is also uneven: many papers show only a single run without a fixed random seed, omit confidence intervals across streams or events, and tune thresholds with information from the test stream (i.e., leakage). Together, these practices undermine fair comparison. Third, there is no shared concept drift benchmark suite for image data streams. Many studies convert static datasets into streams with drift injection (brightness/blur/jitter, domain swaps, class imbalance), but the severities, durations (abrupt vs. gradual), and frequencies are not standardized. This makes evaluations difficult and often under-represents realistic shifts such as sensor aging, seasonal illumination, or camera pose drift.

Table 6 Summary of concept drift detection methods for image data

Method	Strengths	Limitation
MRMR (Li et al. 2023)	Handle multi-label data streams, fast and efficient	Rely on ground-truth labels
CA-Drift (Cobb and Van Looveren 2022)	Allowing a context variable to account for features that are expected to change, provides a more robust application	Do not address problems specific to image data
CDASC (Yuan et al. 2024)	Integrate diffusion learning to improve feature representation, work under scarce labels	Mainly consider random sampling methods, not robust against noise
CDDBS (Okawa and Kobayashi 2021)	Highly responsive for concept drift scenarios	Costly when training multiple models for each class, especially when the number of classes is high
AMSC (Zhang et al. 2018)	Utilize matrix sketching to produce low-dimension representation of data that allows more efficient drift detection	Do not address problems specific to image data
DSM (Li et al. 2024)	Target industrial image characteristics	May not work well on other image datasets
DriftLens (Greco et al. 2024)	Provide a framework for detecting concept drift in unstructured data, fast runtime	Do not address problems specific to image data, may not work well with imbalanced data
CDCSDE (Xu and Klabjan 2021)	Offer flexibility to handle various causes of drift, work under the lag of labels setting	Reliance on multiple metrics may hinder computational efficiency, do not address problems specific to image data
IBDD (Souza et al. 2020)	Fast and suitable for real-time detection	Directly use of raw image pixels may not capture essential semantic features, resulting in reduced detection accuracy
Image Drift (Fuccellaro et al. 2024)	Fast and suitable for real-time detection, work with flexible number of dimensions	Gaussians data distribution assumption, directly use of raw image pixels may trade off accuracy
Visual-Drift (Tran et al. 2024)	Semantic feature extraction and dimension reduction enhances detection accuracy and runtime	Do not support training for cross-domain adaptation
CSI (Tack et al. 2020)	Can detect novel classes in both labeled and unlabeled settings, simple yet effective	The choice of appropriate shifting transformations may hinder the method's accuracy
ORCA (Cao et al. 2022)	Robust to the number of novel classes, handle imbalanced class distributions and generalize to other domains	Require all unlabeled data at once (not online/streaming), model complexity due to multiple classification heads
GCD (Vaze et al. 2022)	Contrastive learning improves feature representation, can estimate the number of novel classes and generalize to different domains	Not designed for online/streaming data, performance depends on quality of clustering

5.2.2 Quantitative comparison comparison

Tables 7 and 8 summarize reported results of concept drift detection methods on virtual and real concept drift, as well as on novel class detection. It is important to note that the results are drawn from different metrics and experimental protocols as reported in the original papers. Consequently, the tables should be read as indicative comparisons rather than as outcomes of a uniform benchmark. Nevertheless, several insights can be highlighted.

Table 7 Results of concept drift detection methods on virtual drift and real concept drift

Method	Setting	Dataset	Results						
			FA	Delay	TPR	F1	ACC	AUC	Others
MRMR (Li et al. 2023)	Not mentioned	Corel16k010 (Li and Wang 2008)	0	0	–	–	–	–	Detection: 68 Missing: 2
		NUS-WIDE (Chua et al. 2009)	9	18	–	–	–	–	Detection: 808 Missing: 11
CDASC (Yuan et al. 2024)	Synthetic abrupt virtual drift by permuting or rotating images.	P.MNIST (Goodfellow et al. 2014)	–	–	–	–	–	–	–
		R.MNIST (Lopez-Paz and Ranzato 2017)	–	–	–	0.815	–	–	HN: 784
CDDBS (Okawa and Kobayashi 2021)	Synthetic virtual drift by using image manipulation	CIFAR10/100 (Krizhevsky and Hinton 2009)	3.42	2.37	–	–	–	–	Acc. Degradn.: 2.43
DSM (Li et al. 2024)	Gradual virtual drift by changes in brightness of the image	RSDDs (Gan et al. 2017)	–	–	–	–	–	0.985	FPR95: 7.90
	Abrupt virtual drift by changes in brightness of the image	RSDDs (Gan et al. 2017)	–	–	–	–	–	0.999	FPR95: 0.17
Drift-Lens (Greco et al. 2024)	Synthetic virtual drift by blurring	STL-10 (Coates et al. 2011)	–	–	–	0.90	–	–	–
CDC-SDE (Xu and Klabjan 2021)	Abrupt virtual drift by introducing USPS into MNIST	MNIST (Deng 2012), USPS (Hull 1994)	–	–	–	–	0.974	–	MTFA(sec.): 252.5 MTD(sec.): 26.3 MDR: 11.1 TD: 10
	Abrupt and gradual virtual drift by introducing USPS into MNIST	MNIST (Deng 2012), USPS (Hull 1994)	–	–	–	–	0.98	–	MTFA(sec.): 178.3 MTD(sec.): 18.3 MDR: 11.1 TD: 11
IBDD (Souza et al. 2020)	Incremental virtual drift by sorting the arrival order of the examples according to the patient's age	Ham10000 (Tschandl et al. 2018)	–	–	–	–	0.778	–	Runtime(sec.): 2131

Table 7 (continued)

Method	Setting	Dataset	Results						
			FA	Delay	TPR	F1	ACC	AUC	Others
Image Drift (Fuccellaro et al. 2024)	Synthetic real concept drift.	CIFAR10/100 (Krizhevsky and Hinton 2009)	–	–	0.10	–	–	–	–
		MNIST (Deng 2012)	–	–	0.40	–	–	–	–
		Fashion-MNIST (Xiao et al. 2017)	–	–	0.80	–	–	–	–

FA: number of False Alarms, Delay: number of Detection Delays, TPR: True Positive Rate or Recall, F1: F1 Score, ACC: Accuracy, AUC: Area Under the Curve, HN: number of Hidden Nodes, Acc. Degradn.: Accuracy Degradation, FPR95: False Positive Rate at 95% True Positive Rate, MTFA: Mean Time between False Alarms, MTD: Mean Time to Detection, MDR: Missed Detection Rate, TD: Total number of Detection

For methods summarized in Table 7, supervised MRMR achieves high robustness with relatively low false alarms and detection misses, though at the cost of requiring ground-truth labels. Among unsupervised methods, CDDBS reports competitive detection rates but suffers from accuracy degradation and scalability issues with large class sets. DSM demonstrates excellent performance in domain-specific industrial streams, with near-perfect AUC under both gradual and abrupt drift, though its applicability beyond industrial settings remains unclear. Methods that operate directly on raw data, such as IBDD and Image Drift, achieve fast runtimes. However, their reliance on pixel-level comparisons limits their ability to capture semantic changes, as shown by poor performance on more complex datasets such as CIFAR. On the other hand, DriftLens, which leverages pretrained features, consistently improves F1 scores, illustrating the benefit of combining deep representation learning with drift detection. Ensemble approaches such as CDCSDE show robustness across abrupt and gradual drifts, but their reliance on multiple metrics incurs significant computational cost. Taken together, these results suggest a trade-off between speed versus semantic sensitivity, with representation learning approaches offering better generalization across diverse settings, while raw pixel methods excel in scenarios requiring real-time efficiency.

In the context of novel class detection (Table 8), the setup typically involves splitting datasets into seen (known) classes and novel (previously unseen) classes. In cases where the same dataset is used for both, classes are partitioned into known and novel subsets; when different datasets are used, one serves as the known domain and the other introduces new classes. Even if the setup is similar, splitting rates and protocols vary across papers, and the table reports results as provided by the original authors.

Several trends emerge. Visual Drift and CSI achieve strong results by leveraging pre-trained representations GCD, which combines self-supervised ViTs with clustering, also generalizes well across datasets, though its accuracy drops on more fine-grained datasets (e.g., CIFAR100, ImageNet). ORCA provides robust accuracy on CIFAR10 but performance deteriorates on more complex datasets such as CIFAR100, revealing sensitivity to dataset complexity and training protocol. DriftLens and AMSC, although not specifically designed for novel classes, report strong F1 scores on STL-10 and MNIST, respectively, showing that representation-focused approaches can extend to novelty detection. Overall, novelty detection methods clearly benefit from representation learning and clustering strate-

Table 8 Results of concept drift detection methods on novel classes

Method	Seen dataset	Novel dataset	Results				
			P	R	F1	ACC	AUC
CA-Drift (Cobb and Van Looveren 2022)	ImageNet (Deng et al. 2009)	ImageNet (Deng et al. 2009)	–	–	–	–	–
AMSC (Zhang et al. 2018)	MNIST (Deng 2012)	MNIST (Deng 2012)	–	–	0.859	0.865	–
DriftLens (Greco et al. 2024)	Intel Image (Rahimzadeh et al. 2021)	Intel Image (Rahimzadeh et al. 2021)	–	–	0.90	–	–
	STL-10 (Coates et al. 2011)	STL-10 (Coates et al. 2011)	–	–	0.96	–	–
Visual-Drift (Tran et al. 2024)	CIFAR10 (Krizhevsky and Hinton 2009)	CIFAR100 (Krizhevsky and Hinton 2009)	0.947	0.903	0.924	–	–
CSI (Tack et al. 2020)	CIFAR10 (Krizhevsky and Hinton 2009)	CIFAR100 (Krizhevsky and Hinton 2009)	–	–	–	–	0.922
	CIFAR10 (Krizhevsky and Hinton 2009)	ImageNet (Deng et al. 2009)	–	–	–	–	0.94
ORCA (Cao et al. 2022)	CIFAR10 (Krizhevsky and Hinton 2009)	CIFAR10 (Krizhevsky and Hinton 2009)	–	–	–	0.904	–
	CIFAR100 (Krizhevsky and Hinton 2009)	CIFAR100 (Krizhevsky and Hinton 2009)	–	–	–	0.43	–
	ImageNet (Deng et al. 2009)	ImageNet (Deng et al. 2009)	–	–	–	0.721	–
	Single-cell (Almanzar et al. 2020)	Single-cell (Almanzar et al. 2020)	–	–	–	0.652	–
GCD (Vaze et al. 2022)	CIFAR10 (Krizhevsky and Hinton 2009)	CIFAR10 (Krizhevsky and Hinton 2009)	–	–	–	0.882	–
	CIFAR100 (Krizhevsky and Hinton 2009)	CIFAR100 (Krizhevsky and Hinton 2009)	–	–	–	0.665	–
	ImageNet (Deng et al. 2009)	ImageNet (Deng et al. 2009)	–	–	–	0.663	–

P: Precision, R: Recall, F1: F1 Score, ACC: Accuracy, AUC: Area Under the Curve

gies, which enable separation of known and novel categories. However, their performance varies significantly across datasets, suggesting that scalability and adaptability remain open challenges.

In summary, while virtual and real drift detection methods differ widely in their evaluation settings, a clear distinction can be made: methods relying on representation learning offer improved accuracy and robustness, while raw image-based approaches prioritize efficiency. In contrast, novelty detection methods form a distinct family that primarily focuses on robust representation learning, but their generalization across datasets remains an active challenge.

5.2.3 Strengths and limitations

Among the surveyed methods, MRMR is the only one that fully relies on ground-truth labels, making it a supervised approach. While its reliance on labels makes the method less practical in real-world scenarios where annotation is costly and time-consuming, MRMR demonstrates that combining label distributions with image feature distributions can significantly improve both speed and robustness, especially in multi-label data streams. This illustrates a broader trade-off in drift detection: accuracy and robustness gained from supervision versus scalability and practicality in unlabeled settings.

In contrast, most existing approaches are unsupervised, yet they differ in how they represent image data and in the type of drifts they target. For example, CA-Drift, AMSC, and CDDBS extend classical statistical or sketching-based strategies. While these methods introduce novel ideas—such as contextual variables in CA-Drift or matrix sketching in AMSC—they remain largely agnostic to the semantic content of images, which limits their ability to capture subtle image-driven changes. Moreover, computational scalability remains a bottleneck, particularly for CDDBS when applied to high-class regimes. In comparison, CDASC and DSM begin to address image-specific challenges by integrating diffusion learning and domain-tailored modeling (industrial image streams, in the case of DSM). These methods highlight a growing shift in the field toward improved feature representations as the foundation for effective drift detection.

Another family of works attempts to bypass feature extraction altogether by operating directly on raw image data. Notable examples include IBDD and Image Drift. These approaches are extremely fast, making them suitable for real-time deployment. However, their reliance on pixel-level comparisons limits their ability to capture semantic changes in image content. Furthermore, Image Drift assumes Gaussianity in the data distribution, an assumption that rarely holds in practice. This reflects a recurring trade-off: methods optimized for speed often sacrifice semantic awareness, restricting their applicability to more complex domains.

A more recent trend leverages pretrained deep models as feature extractors. DriftLens and Visual Drift follow this paradigm by combining deep feature extraction with dimension reduction. This hybrid strategy improves both accuracy and computational efficiency, demonstrating strong potential for general-purpose use. However, neither method supports domain adaptation or addressing core image problems, limiting their applicability in cross-domain or transfer scenarios—an increasingly important challenge in real-world settings.

Recently, a distinct family of methods has emerged that focuses on novelty detection, i.e., the identification of previously unseen classes. In contrast to approaches that monitor distributional shifts within a fixed label set, novelty detection methods explicitly address structural changes in the label space. As this has become a highly active research direction, it is essential to distinguish these approaches from traditional drift detection. Most methods in this category emphasize representation learning using self-supervised methods, aiming to build feature spaces that can separate known classes while remaining flexible enough to accommodate novel ones.

Representative examples include CSI, which leverages contrastive learning to build robust feature spaces and detect both calibration shifts and novel classes. ORCA jointly learns to classify known categories while discovering new ones via multiple classification heads and pairwise similarity objectives, making it powerful in offline discovery scenarios but less suitable for streaming due to its reliance on full access to unlabeled data. GCD advances this direction by combining self-supervised Vision Transformers (DINO) (Caron et al. 2021) with semi-supervised clustering, enabling strong cross-domain generalization but lacking online adaptability.

Taken together, these methods demonstrate that the emphasis on self-supervised learning and clustering help with detecting novel classes, one of the causes of concept drift. Overall, self-supervised pretraining provides stronger, more generalizable representations that reduce sensitivity to small and moderate drifts, especially in virtual drifts and unseen-but-similar domains. However, for severe concept drift or novel class emergence, representation learning alone is insufficient—continual adaptation or novelty detection mechanisms are still necessary.

6 Challenges and future research direction

This survey has presented a comprehensive overview of concept drift detection approaches for image data, beginning with a broad discussion of the concept drift domain and narrowing down to a critical analysis of existing detection methods specifically for image data. We examined key criteria that define an effective drift detection method, highlighting their strengths, limitations, and applicability. Additionally, we explored essential aspects such as datasets and evaluation metrics, which play a crucial role in assessing detection performance. Based on our analysis, we identify several key challenges in image-based concept drift detection along with promising future research directions. A detailed discussion of these challenges is provided below, while Table 9 summarizes the current progress and remaining open issues in a concise overview.

Table 9 Overview of solved and unsolved challenges in image-based concept drift detection

Area	Progress/solved aspects	Open challenges/future directions
Generalization	Initial methods (Tran et al. 2024; Greco et al. 2024) show success in generalizing in specific domains using pre-trained deep learning models (e.g., natural images, small benchmarks)	Poor cross-domain transfer (e.g., medical, remote sensing); most focus only on virtual drift or novel classes, neglecting in-class evolution
Feature Learning	Contrastive/self-supervised methods (Tack et al. 2020; Cao et al. 2022; Vaze et al. 2022) provide robust semantic embeddings for novelty detection	Most assume offline or transductive settings; limited adaptation for online/streaming scenarios; shallow CNNs still dominate CDD literature
Computational Efficiency	Pixel-level or lightweight approaches reduce feature extraction cost (Fuccellaro et al. 2024; Souza et al. 2020)	Trade-off between speed and accuracy; need lightweight yet semantically rich feature extractors; incremental and modular updating strategies remain underexplored
Evaluation Metrics	Introduction of drift-specific metrics (e.g., Detection Delay, False Alarms, Missed Detections)	Lack of standardization across studies; metrics often inconsistently reported or omitted; need real-time evaluation protocols
Benchmarking	Synthetic drift scenarios (noise, occlusions, class addition) widely used; Semantic Shift Benchmark provides first step toward standardization	Absence of widely accepted benchmark datasets with diverse, annotated drift types (virtual drift, novel classes, in-class evolution), and lack of real-world image stream datasets

6.1 Generalization

One of the central challenges in concept drift detection for image data is achieving generalization across diverse domains and drift scenarios.

Across Different Domains Many existing methods are designed with assumptions tied to specific datasets or visual domains. This limits their ability to generalize to new application areas such as remote sensing, medical imaging, or industrial inspection. Methods tailored to natural images often underperform when transferred to domains with different statistical properties, such as satellite imagery with multi-spectral bands.

Across different causes of drift Most image-based concept drift detection methods are tailored to a specific cause of drift, either virtual drift (Yuan et al. 2024; Okawa and Kobayashi 2021; Li et al. 2024; Souza et al. 2020) or real concept drift (Cobb and Van Looveren 2022; Zhang et al. 2018; Tran et al. 2024). Moreover, within real drift, the majority target only the emergence of novel classes while largely neglecting in-class evolution—a more subtle and underexplored form of drift. Future work should aim to develop methods capable of identifying and adapting to multiple types of drift simultaneously.

6.2 Feature learning for image data

Effective feature learning is essential for improving drift detection performance, yet remains underdeveloped in many existing methods for image data. Detecting concept drift in images requires not only monitoring changes in distributions but also understanding semantic shifts that occur over time—something that heavily depends on the quality of the learned feature representations.

Most concept drift detection methods applied to image data rely on simplistic feature extraction techniques, often using shallow CNN architectures (e.g., two-layer CNNs) (Zhang et al. 2018; Yuan et al. 2024; Xu and Klabjan 2021). While these approaches provide a basic representation of images, they may not be sufficient for capturing the complex, hierarchical features needed to distinguish subtle distribution shifts.

Recent advances from the computer vision community—such as CSI (Tack et al. 2020), ORCA (Cao et al. 2022), and GCD (Vaze et al. 2022)—demonstrate the effectiveness of contrastive learning for producing robust and semantically meaningful embeddings. These methods employ self-supervised techniques (e.g., SimCLR (Chen et al. 2020), DINO (Caron et al. 2021)) or pairwise similarity learning to structure the feature space, enabling better discrimination between seen and novel classes. However, these methods often assume full access to unlabeled data, operating in a transductive or batch-mode setting. This is fundamentally different from streaming or online environments, where concept drift often occurs and where data arrives sequentially. A further research direction should involve adapting contrastive and self-supervised representation learning to streaming settings.

6.3 Computational efficiency

Scalability and real-time performance remain significant challenges in image-based concept drift detection. The inherently high dimensionality and unstructured nature of image data lead to increased computational and memory demands, particularly when processing large-scale or high-frequency data streams.

While some recent methods have explored computationally efficient alternatives that operate directly on raw image pixels to avoid expensive feature extraction (Fuccellaro et al. 2024; Souza et al. 2020), these approaches often involve a trade-off between speed and detection accuracy. Such methods may sacrifice semantic richness from feature extraction, which is critical for identifying subtle or high-level distributional changes in complex image streams.

To address these limitations, future research should prioritize the following directions:

- *Lightweight feature extraction and dimensionality reduction* Investigating methods such as adaptive PCA, sparsity-inducing neural architectures, or compressed representations that can reduce computation while preserving key semantic information.
- *Modular and incrementally updatable frameworks* Designing flexible architectures that support partial model updates, on-the-fly adaptation, or streamlined retraining, thereby avoiding the need to recompute the entire pipeline after each drift detection or adaptation step.

6.4 Evaluation metrics and benchmarking

Evaluation Metrics The evaluation of concept drift detection methods for image data currently suffers from a lack of standardization, making fair comparison and reproducibility across studies difficult. Different works employ different metrics depending on their specific assumptions and settings, often without justification or consistency. While common drift-specific metrics such as Detection Delay, False Alarms, and Missed Detection Rate are used, they are frequently reported inconsistently or omitted altogether. For unsupervised methods, performance is often assessed indirectly through proxy evaluations, such as downstream classification accuracy or clustering purity, which may not fully reflect drift detection capabilities.

Benchmarking There is a lack of shared benchmark datasets that represent a variety of drift types and define a problem setting. Most current studies simulate drift via manipulations like noise, occlusions, or new class introduction, often inconsistently. A potential new benchmark should include a range of predefined drift scenarios (e.g., virtual drift, novel class emergence, in-class evolution) and clearly annotated ground-truth change points, similar to the design of the Semantic Shift Benchmark (Vaze et al. 2022).

To advance the field, it is crucial for the community to establish standardized evaluation protocols and, equally importantly, to develop shared benchmarks based on annotated, real-world image streams. Such benchmarks would enable systematic comparison across methods and ensure that evaluation settings reflect the practical challenges of deploying concept drift detection in real-world applications.

Table 10 Datasets

Dataset	Size	Resolution	No. Classes	Description	Used by references
ImageNet (Deng et al. 2009)	1,281,167	Varied	1000	Images in RGB format with varying resolutions and categories	(Cobb and Van Looveren 2022)
CIFAR10 (Krizhevsky and Hinton 2009)	60,000	32 × 32	10	Images of common objects: airplanes, cars, birds, cats,..	(Okawa and Kobayashi 2021)
STL-10 (Coates et al. 2011)	113,000	96 × 96	10	Inspired by CIFAR10, fewer labeled but more unlabeled samples	(Greco et al. 2024)
MNIST (Deng 2012)	70,000	28 × 28	10	Images of handwritten digits from 0 to 10	(Yuan et al. 2024; Xu and Klabjan 2021; Zhang et al. 2018)
USPS (Hull 1994)	9298	16×16	10	Images of digits scanned from envelopes by the U.S. Postal Service	(Xu and Klabjan 2021)
Intel Image (Rahimzadeh et al. 2021)	25,000	150 × 150	6	Images of natural scenes around the world	(Greco et al. 2024)
Ham10000 (Tschandl et al. 2018)	10,015	600 × 450	7	Dermatoscopic images specifically designed for skin lesion analysis	(Souza et al. 2020)
RSDD (Gan et al. 2017)	5700	256 × 256	2	Images used for detecting surface defects on rail tracks	(Li et al. 2024)

Appendix A: Datasets

A range of existing image datasets has been applied in concept drift detection research, though many were not originally designed for this purpose. Datasets like ImageNet (Deng et al. 2009), CIFAR-10 (Krizhevsky and Hinton 2009), STL-10 (Coates et al. 2011), MNIST (Deng 2012), and USPS (Hull 1994) are widely used due to their extensive image collections, allowing researchers to simulate temporal changes and class evolution. These datasets typically consist of diverse categories, making them suitable for testing class distribution shifts in incremental learning contexts. More domain-specific datasets are also used, such as Ham10000 (Tschandl et al. 2018), which includes skin lesion images for medical applications, or RSDDs (Gan et al. 2017), focusing on surface defect detection in industrial settings, or Intel Image (Rahimzadeh et al. 2021), including images of natural scenes around the world. Table 10 provides a comprehensive summary of all datasets employed in the surveyed studies discussed in Sect. 4.

In concept drift research, synthetic datasets have played a crucial role due to their controlled environments, allowing researchers to systematically simulate different types of drift, such as gradual, sudden, or recurring. Synthetic drift scenarios enable precise manipulations, such as introducing new classes over time or applying various image manipulation techniques to mimic environmental changes. Common manipulations (Okawa and Kobayashi 2021) include (a) GaussianBlur (Blur) to simulate degradation in image clarity, (b) Darken to mimic lighting changes, (c) Rotate to replicate shifts in object orientation, (d)

GaussianNoise (Noisy) to introduce random interference, (e) Rain to add weather effects, and (f) Snow for seasonal variations. These alterations help create controlled drift scenarios, which can be arranged in various ways to simulate gradual changes, abrupt shifts, or recurring drift events. An example of this approach is the Rotated MNIST (R.MNIST) (Lopez-Paz and Ranzato 2017) and Permuted MNIST (P.MNIST) (Goodfellow et al. 2014) datasets, which apply specific transformations to the original MNIST dataset to simulate drift. In R.MNIST, digits are incrementally rotated, creating a continuous drift as the angle of rotation changes. In P.MNIST, pixels are permuted randomly, simulating a drift that introduces new patterns over time.

Real-world datasets, while more complex and less predictable, offer a more authentic representation of drift as it naturally occurs. However, given that concept drift detection in image data is still an emerging field, most current research relies on synthetic datasets. Real-world datasets for drift detection are rare, particularly in image-based applications, and lack the explicit drift labeling or temporal annotations often needed for detailed analysis.

Overall, currently available datasets in image data for concept drift detection face several challenges. First, there is a lack of diversity in drift types across available datasets; most datasets primarily support simulated or incremental changes, with few options that capture complex or overlapping drift patterns seen in real-world settings. Secondly, acquiring large-scale datasets that cover long-term temporal changes is challenging, making it difficult to assess how well drift detection methods perform over extended periods. Addressing these limitations is critical to advancing the field, as a more diverse set of datasets would allow for more robust benchmarking and evaluation.

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Declarations

Conflict of interest The authors declare no Conflict of interest.

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